David Jones Marketing Campaign Strategy: Analytics and Database Design Report
Assignment 1

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BUSA8090 - Data and Visualisation for Business

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Part A – Database Design and Development

1.1 Entity Identification

Entity Name	Entity Description
Platform	It represents the social media platforms where advertisements are published such as Instagram, Facebook, X, TikTok and YouTube,
Campaign	It represents a marketing campaign which is run by David Jones such as name, budget and actives date
Advertisement	It represents an individual advertisement placed ono an social media as a part of a campaign
Customer	It represents a registered online shopper who may click on advertisement and place order
Click	It records customer clicking on an advertisement which capture when and which ad was clicked
Order	It represents order placed by a customer on the David Jones
Product	It represents a product available to purchase on David Jones online store
Contain	It is an associative entity that resolves many to many relationships between order and product while recording the quantity of each product in an order.

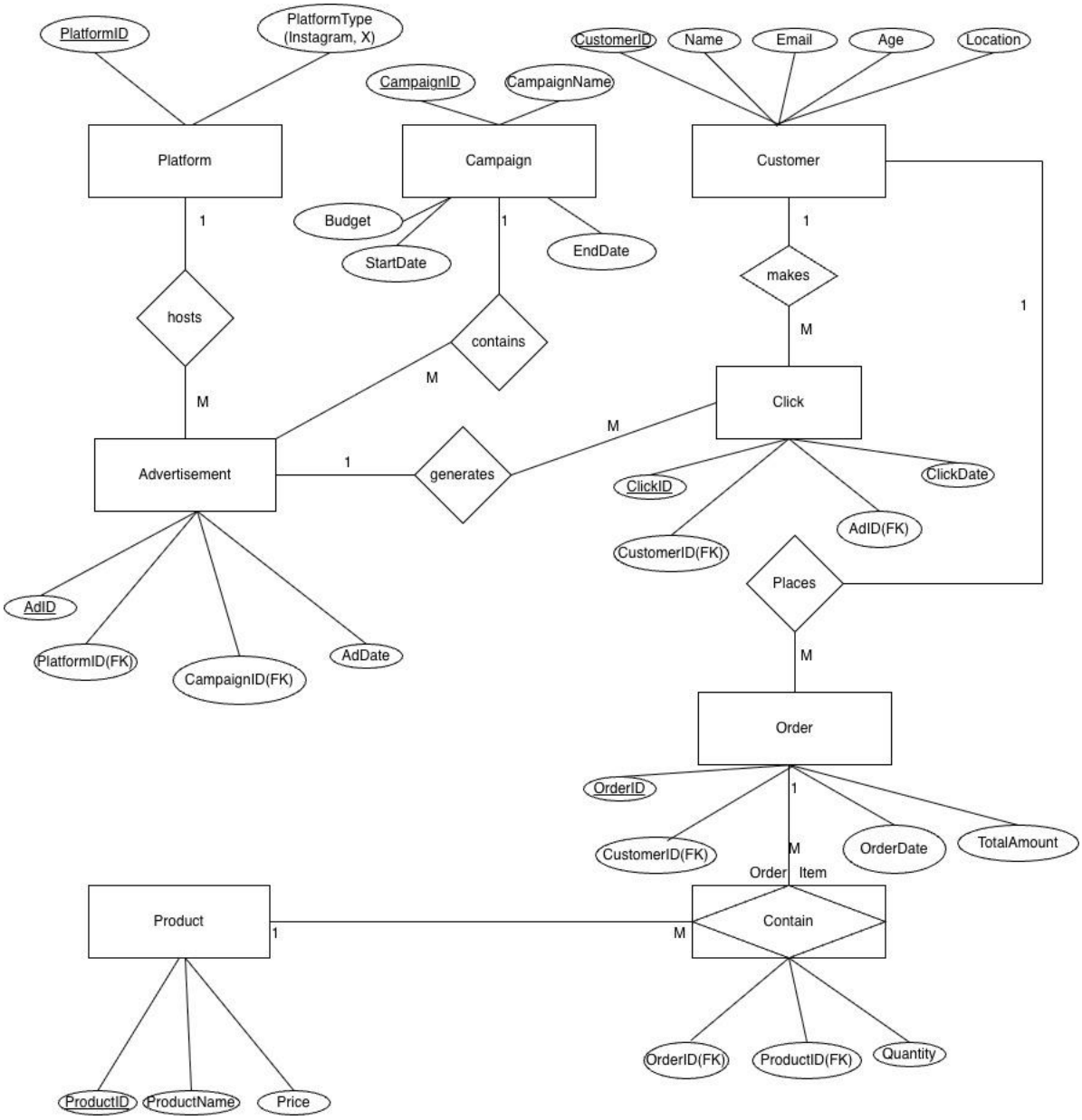
1.2 Identify Attributes, Primary Keys and Foreign Keys

Entity Name	Attributes
Platform	PlatformID (PK), PlatformType
Campaign	CampaignID(PK), CampaignName, Budget, StartDate, EndDate
Advertisement	AdId(PK), PlatformID (FK), CampaignID(FK), AdDate
Customer	CustomerId(PK), Name, Email,Age, Location
Click	ClickID(PK), CustomerID(FK), AdId(FK), ClickDate
Order	OrderID(PK), CustomerID(FK), OrderDate, TotalAmount
Contain	OrderID (PK, FK), ProductID (PK, FK), Quantity
Product	ProductID(PK), ProductName, Price

1.3 Relationship Definition and Cardinality Analysis

Relationship	Entities Involved	Cardinality	Description
hosts	Platform → Advertisement	One – to-Many(1:M)	One platform hosts many advertisements. Each advertisement is published on exactly one platform
contains	Campaign → Advertisement	One – to – Many (1:M)	One campaign contains many advertisements. Each advertisement belongs to exactly one campaign
makes	Customer → Click	One-to-Many(1:M)	One customer can make many clicks on advertisements. Each click is made by exactly one customer
generates	Advertisement → Click	One-to-Many(1:M)	One advertisement can generate many clicks. Each click is linked to exactly one advertisement
places	Customer → Order	One-to-Many(1:M)	One customer can place many orders. Each order belongs to exactly one customer
Order Item	Order ↔ Product via Contain	Many-to-Many(M:M)	One order can contain many products, and one product can appear in many orders. The contains associative entity resolves this relationship using a composite primary key

1.4 ERD diagram



1.5 Table implementation

The following SQL statements were used to create all tables in MySQL workbench. Tables were corrected in the correct order to satisfy key dependencies.

```sql
CREATE DATABASE IF NOT EXISTS david_jones_db;
USE david_jones_db;

CREATE TABLE Platform (
    PlatformID      INT          PRIMARY KEY,
    PlatformType    VARCHAR(50)  NOT NULL
);

CREATE TABLE Campaign (
    CampaignID      INT          PRIMARY KEY,
    CampaignName    VARCHAR(100) NOT NULL,
    Budget          DECIMAL(10,2),
    StartDate       DATE,
    EndDate         DATE
);

CREATE TABLE Advertisement (
    AdID            INT          PRIMARY KEY,
    PlatformID      INT,
    CampaignID      INT,
    AdDate          DATE,
    FOREIGN KEY (PlatformID)  REFERENCES Platform(PlatformID),
    FOREIGN KEY (CampaignID)  REFERENCES Campaign(CampaignID)
);

CREATE TABLE Customer (
    CustomerID      INT          PRIMARY KEY,
    Name            VARCHAR(100) NOT NULL,
    Email           VARCHAR(100),
    Age             INT,
    Location        VARCHAR(100)
);
```

```sql
CREATE TABLE Click (
    ClickID         INT          PRIMARY KEY,
    CustomerID      INT,
    AdID            INT,
    ClickDate       DATE,
    FOREIGN KEY (CustomerID)  REFERENCES Customer(CustomerID),
    FOREIGN KEY (AdID)        REFERENCES Advertisement(AdID)
);

CREATE TABLE `Order` (
    OrderID         INT          PRIMARY KEY,
    CustomerID      INT,
    OrderDate       DATE,
    TotalAmount     DECIMAL(10,2),
    FOREIGN KEY (CustomerID)  REFERENCES Customer(CustomerID)
);

CREATE TABLE Product (
    ProductID       INT          PRIMARY KEY,
    ProductName     VARCHAR(100) NOT NULL,
    Price           DECIMAL(10,2)
);

CREATE TABLE Contain (
    OrderID         INT,
    ProductID       INT,
    Quantity        INT,
    PRIMARY KEY (OrderID, ProductID),
    FOREIGN KEY (OrderID)    REFERENCES `Order`(OrderID),
    FOREIGN KEY (ProductID)  REFERENCES Product(ProductID)
);
```

1.6 Dummy Data insertion

All the tables were filled with the following sample data to support data retrieval and business analytics. All the key entities have between 10 and 30 rows to guarantee meaningful and diverse analytics result

Classified

```sql
-- Platform
INSERT INTO Platform (PlatformID, PlatformType)
VALUES
(1,  'Instagram'),
(2,  'Facebook'),
(3,  'X'),
(4,  'TikTok'),
(5,  'YouTube'),
(6,  'Pinterest'),
(7,  'Snapchat'),
(8,  'LinkedIn'),
(9,  'Reddit'),
(10, 'Google Ads');

-- Campaign
INSERT INTO Campaign (CampaignID, CampaignName, Budget, StartDate, EndDate)
VALUES
(1,  'EOFY Sale 2025',           50000.00, '2025-06-01', '2025-06-30'),
(2,  'Winter Warmup 2025',       30000.00, '2025-07-01', '2025-07-31'),
(3,  'Spring Collection 2025',   25000.00, '2025-09-01', '2025-09-30'),
(4,  'Black Friday 2025',        60000.00, '2025-11-25', '2025-11-30'),
(5,  'Christmas Sale 2025',      45000.00, '2025-12-01', '2025-12-25'),
(6,  'Valentines Day 2025',      20000.00, '2025-02-01', '2025-02-14'),
(7,  'Easter Sale 2025',         22000.00, '2025-04-01', '2025-04-20'),
(8,  'Mothers Day 2025',         18000.00, '2025-05-01', '2025-05-11'),
(9,  'Fathers Day 2025',         18000.00, '2025-08-01', '2025-09-07'),
(10, 'Back to School 2025',      15000.00, '2025-01-15', '2025-02-01'),
(11, 'New Year New You 2025',    12000.00, '2025-01-01', '2025-01-14'),
(12, 'Melbourne Cup Campaign',   10000.00, '2025-10-25', '2025-11-04');

-- Advertisement
INSERT INTO Advertisement (AdID, PlatformID, CampaignID, AdDate)
VALUES
(1,  1,  1,  '2025-06-01'),
(2,  2,  1,  '2025-06-02'),
(3,  3,  1,  '2025-06-03'),
(4,  4,  1,  '2025-06-04'),
(5,  5,  2,  '2025-07-01'),
(6,  1,  2,  '2025-07-02'),
(7,  2,  2,  '2025-07-05'),
(8,  3,  3,  '2025-09-01'),
(9,  4,  3,  '2025-09-03'),
(10, 6,  3,  '2025-09-05'),
(11, 1,  4,  '2025-11-25'),
(12, 2,  4,  '2025-11-26'),
(13, 3,  4,  '2025-11-27'),
(14, 4,  5,  '2025-12-01'),
(15, 5,  5,  '2025-12-02'),
(16, 6,  6,  '2025-02-01'),
(17, 7,  6,  '2025-02-05'),
(18, 1,  7,  '2025-04-01'),
(19, 2,  7,  '2025-04-05'),
(20, 8,  8,  '2025-05-01'),
(21, 9,  9,  '2025-08-15'),
(22, 10, 9,  '2025-08-20'),
(23, 1,  10, '2025-01-15'),
(24, 3,  11, '2025-01-01'),
(25, 5,  12, '2025-10-25');


-- Product
INSERT INTO Product (ProductID, ProductName, Price)
VALUES
(4001, 'Winter Jacket',  189.99),
(4002, 'Dress Shirt',     79.99),
(4003, 'Sneakers',       129.99),
(4004, 'Handbag',        249.99),
(4005, 'Sunglasses',      89.99),
(4006, 'Perfume',        119.99),
(4007, 'Wool Coat',      299.99),
(4008, 'Silk Scarf',      59.99),
(4009, 'Leather Belt',    49.99),
(4010, 'Running Shoes',  149.99),
(4011, 'Casual Dress',   109.99),
(4012, 'Denim Jeans',     99.99),
(4013, 'Polo Shirt',      69.99),
(4014, 'Ankle Boots',    179.99),
(4015, 'Gym Bag',         79.99);
```

Classified

```sql
-- Customer
INSERT INTO Customer (CustomerID, Name, Email, Age, Location)
VALUES
(1001, 'Emma Wilson',      'emma.wilson@gmail.com',      28, 'Sydney'),
(1002, 'James Brown',      'james.brown@gmail.com',      35, 'Melbourne'),
(1003, 'Olivia Smith',     'olivia.smith@yahoo.com',     22, 'Brisbane'),
(1004, 'Liam Johnson',     'liam.johnson@gmail.com',     45, 'Perth'),
(1005, 'Sophia Davis',     'sophia.davis@outlook.com',   31, 'Sydney'),
(1006, 'Noah Taylor',      'noah.taylor@gmail.com',      27, 'Adelaide'),
(1007, 'Ava Martinez',     'ava.martinez@hotmail.com',   33, 'Sydney'),
(1008, 'William Lee',      'william.lee@gmail.com',      29, 'Melbourne'),
(1009, 'Isabella White',   'isabella.white@yahoo.com',   24, 'Brisbane'),
(1010, 'Mason Harris',     'mason.harris@gmail.com',     38, 'Sydney'),
(1011, 'Mia Clark',        'mia.clark@outlook.com',      26, 'Perth'),
(1012, 'Ethan Lewis',      'ethan.lewis@gmail.com',      41, 'Melbourne'),
(1013, 'Charlotte Walker', 'charlotte.walker@yahoo.com', 23, 'Sydney'),
(1014, 'Alexander Hall',   'alex.hall@gmail.com',        36, 'Adelaide'),
(1015, 'Amelia Young',     'amelia.young@hotmail.com',   30, 'Brisbane'),
(1016, 'Henry King',       'henry.king@gmail.com',       44, 'Sydney'),
(1017, 'Harper Scott',     'harper.scott@yahoo.com',     25, 'Melbourne'),
(1018, 'Lucas Green',      'lucas.green@gmail.com',      32, 'Perth'),
(1019, 'Evelyn Adams',     'evelyn.adams@outlook.com',   27, 'Sydney'),
(1020, 'Benjamin Baker',   'ben.baker@gmail.com',        39, 'Brisbane'),
(1021, 'Abigail Nelson',   'abigail.nelson@yahoo.com',   21, 'Sydney'),
(1022, 'Daniel Carter',    'daniel.carter@gmail.com',    34, 'Melbourne'),
(1023, 'Emily Mitchell',   'emily.mitchell@hotmail.com', 28, 'Adelaide'),
(1024, 'Matthew Perez',    'matt.perez@gmail.com',       43, 'Sydney'),
(1025, 'Sofia Roberts',    'sofia.roberts@yahoo.com',    26, 'Perth');
```

```sql
-- Click
INSERT INTO Click (ClickID, CustomerID, AdID, ClickDate)
VALUES
(2001, 1001, 1,  '2025-06-02'),
(2002, 1002, 2,  '2025-06-03'),
(2003, 1003, 3,  '2025-06-04'),
(2004, 1004, 4,  '2025-06-05'),
(2005, 1005, 5,  '2025-07-02'),
(2006, 1006, 6,  '2025-07-03'),
(2007, 1007, 7,  '2025-07-06'),
(2008, 1008, 8,  '2025-09-02'),
(2009, 1009, 9,  '2025-09-04'),
(2010, 1010, 10, '2025-09-06'),
(2011, 1011, 11, '2025-11-25'),
(2012, 1012, 12, '2025-11-26'),
(2013, 1013, 13, '2025-11-27'),
(2014, 1014, 14, '2025-12-02'),
(2015, 1015, 15, '2025-12-03'),
(2016, 1016, 16, '2025-02-02'),
(2017, 1017, 17, '2025-02-06'),
(2018, 1018, 18, '2025-04-02'),
(2019, 1019, 19, '2025-04-06'),
(2020, 1020, 20, '2025-05-02'),
(2021, 1021, 21, '2025-08-16'),
(2022, 1022, 22, '2025-08-21'),
(2023, 1023, 23, '2025-01-16'),
(2024, 1024, 24, '2025-01-02'),
(2025, 1025, 25, '2025-10-26'),
(2026, 1001, 11, '2025-11-26'),
(2027, 1005, 14, '2025-12-02'),
(2028, 1010, 1,  '2025-06-03'),
(2029, 1015, 8,  '2025-09-03'),
(2030, 1020, 18, '2025-04-03');
```

```sql
-- Order
INSERT INTO `Order` (OrderID, CustomerID, OrderDate, TotalAmount)
VALUES
(3001, 1001, '2025-06-02',  250.00),
(3002, 1002, '2025-06-03',  180.00),
(3003, 1003, '2025-06-04',  420.00),
(3004, 1004, '2025-06-05',  310.00),
(3005, 1005, '2025-07-02',   95.00),
(3006, 1006, '2025-07-03',  560.00),
(3007, 1007, '2025-07-06',  200.00),
(3008, 1008, '2025-09-02',  340.00),
(3009, 1009, '2025-09-04',  150.00),
(3010, 1010, '2025-09-06',  890.00),
(3011, 1011, '2025-11-25',  430.00),
(3012, 1012, '2025-11-26',  275.00),
(3013, 1013, '2025-11-27',  615.00),
(3014, 1014, '2025-12-02',  190.00),
(3015, 1015, '2025-12-03',  320.00),
(3016, 1016, '2025-02-02',  480.00),
(3017, 1017, '2025-02-06',  210.00),
(3018, 1018, '2025-04-02',  390.00),
(3019, 1019, '2025-04-06',  750.00),
(3020, 1020, '2025-05-02',  520.00),
(3021, 1021, '2025-08-16',  160.00),
(3022, 1022, '2025-08-21',  445.00),
(3023, 1023, '2025-01-16',  280.00),
(3024, 1024, '2025-01-02',  370.00),
(3025, 1025, '2025-10-26',  495.00),
(3026, 1001, '2025-11-26',  320.00),
(3027, 1005, '2025-12-02',  175.00),
(3028, 1010, '2025-06-03',  640.00),
(3029, 1015, '2025-09-03',  290.00),
(3030, 1020, '2025-04-03',  415.00);
```

```sql
-- Contain (Order Item)
INSERT INTO Contain (OrderID, ProductID, Quantity)
VALUES
(3001, 4001, 1),
(3001, 4002, 2),
(3002, 4003, 1),
(3002, 4013, 1),
(3003, 4004, 1),
(3003, 4005, 2),
(3004, 4001, 1),
(3004, 4006, 1),
(3005, 4005, 1),
(3006, 4004, 2),
(3006, 4007, 1),
(3007, 4002, 1),
(3007, 4008, 2),
(3008, 4003, 2),
(3008, 4006, 1),
(3009, 4009, 1),
(3009, 4010, 1),
(3010, 4007, 1),
(3010, 4004, 1),
(3010, 4011, 2),
(3011, 4012, 2),
(3011, 4013, 1),
(3012, 4002, 1),
(3012, 4014, 1),
(3013, 4004, 1),
(3013, 4006, 2),
(3014, 4013, 2),
(3015, 4010, 1),
(3015, 4015, 1),
(3016, 4007, 1),
(3016, 4008, 1),
(3017, 4011, 1),
(3017, 4009, 2),
```

Table	Rows Inserted	Purpose
Platform	10	Covers all the major social media platforms used by David Jones
Campaign	12	Cover all year seasonal and event based campaigns
Advertisement	25	Ads on all median and campaigns
Customer	25	Various customers in Sydney, Melbourne, Brisbane, Perth, Adelaide
Click	30	Purchase orders made by the customers after clicking on adverts

Order	30	Purchase orders made by the customers after clicking on adverts
Product	15	Range of David Jones fashion and lifestyle products
Order Item	58	Order items are used to connect products to orders and to provide quantities

Part B – Business Analytics Report

1.7 Business concern 1

Name and Description:

Platform Performance: Which social media platform generates the most clicks?

To effectively spend the advertising budget, the CMO must determine the social media platform that brings the greatest number of customer clicks. Investment in platforms that perform poorly is a waste of resources that could be redirected to high-performing channels such as Instagram or Facebook

SQL Query

```sql
-- QUERY 1: Which social media platform generates the most clicks?
SELECT p.PlatformType, COUNT(c.ClickID) AS TotalClicks
FROM Platform p
JOIN Advertisement a ON p.PlatformID = a.PlatformID
JOIN Click c ON a.AdID = c.AdID
GROUP BY p.PlatformType
ORDER BY TotalClicks DESC;
```

SQL Result

PlatformType	TotalClicks
Instagram	8
X	5
Facebook	4
TikTok	4
YouTube	3
Pinterest	2
Snapchat	1
LinkedIn	1
Reddit	1
Google Ads	1

Justification and Insights

This query will be used to combine the Platform, Advertisement and Click tables to get to get a count of total clicks per platform. The findings will provide information on which platforms these social media platforms like Instagram, Facebook or TikTok can be most engaged by customers. The CMO should increase ad spends that have good performances and cut down on the low-click platforms. In the case of EOFY sale, David Jones should consider the use of Instagram and Facebook where they are most likely to find are the fashion audiences are most active.

1.8 Business concern 2

Name and Description:

Campaign Revenue and ROI: Which campaign generated the most revenue?

Some campaigns do not bring the same amount of revenue in proportion to its budget. By determining the campaigns that provide the best returns on investments, the CMO needs to determine the campaign planning and budget allocation across seasonal events.

SQL Query

```sql
-- QUERY 2: Which campaign generated the most revenue?
SELECT ca.CampaignName, ca.Budget, SUM(o.TotalAmount) AS TotalRevenue,
ROUND(SUM(o.TotalAmount) / ca.Budget * 100, 2) AS ROI_Percentage
FROM Campaign ca
JOIN Advertisement a ON ca.CampaignID = a.CampaignID
JOIN Click cl ON a.AdID = cl.AdID
JOIN `Order` o ON cl.CustomerID = o.CustomerID
GROUP BY ca.CampaignName, ca.Budget
ORDER BY TotalRevenue DESC;
```

SQL Result

CampaignName	Budget	TotalRevenue	ROI_Percentage
EOFY Sale 2025	50000.00	3010.00	6.02
Spring Collection 2025	25000.00	2630.00	10.52
Easter Sale 2025	22000.00	2075.00	9.43
Black Friday 2025	60000.00	1890.00	3.15
Christmas Sale 2025	45000.00	1070.00	2.38
Winter Warmup 2025	30000.00	1030.00	3.43
Mothers Day 2025	18000.00	935.00	5.19
Valentines Day 2025	20000.00	690.00	3.45
Fathers Day 2025	18000.00	605.00	3.36
Melbourne Cup Campaign	10000.00	495.00	4.95
New Year New You 2025	12000.00	370.00	3.08
Back to School 2025	15000.00	280.00	1.87

Justification and Insights

This query joins the campaign, Advertisements, Click and Order tables to compute the total revenue per campaign and an ROI percentage. Raising campaigns that have a ROI of less than 100% are failing to recover their costs and should be restructured. To repeat the strategy of high ROI campaign like the EOFY sale and Black Friday, the CMO must replicate the strategy in future palling cycles.

1.9 Business concern 3

Name and Description:

Click through rates: Which platform has the best ad effectiveness?

Raw clicks alone do not provide ad efficiency. A platform that has lower advertisement and high clicks per advert has a higher Click through rate, meaning its audience are more responsive to David Jones advertising content.

SQL Query

```
-- QUERY 3: Which platform has the best click through rate?
SELECT p.PlatformType, COUNT(DISTINCT a.AdID) AS TotalAds,
COUNT(c.ClickID) AS TotalClicks,
ROUND(COUNT(c.ClickID) / COUNT(DISTINCT a.AdID), 2) AS CTR_ClicksPerAd
FROM Platform p
JOIN Advertisement a ON p.PlatformID = a.PlatformID
LEFT JOIN Click c ON a.AdID = c.AdID
GROUP BY p.PlatformType
ORDER BY CTR_ClicksPerAd DESC;
```

SQL Result

PlatformType	TotalAds	TotalClicks	CTR_ClicksPerAd
Instagram	5	8	1.60
TikTok	3	4	1.33
X	4	5	1.25
Facebook	4	4	1.00
Google Ads	1	1	1.00
LinkedIn	1	1	1.00
Pinterest	2	2	1.00
Reddit	1	1	1.00
Snapchat	1	1	1.00
YouTube	3	3	1.00

Justification and Insights

This query consists of a LEFT JOIN as to include all the platforms regardless of whether they have any clicks. The CTR is computed as the total clicks divided by the total distinct ads, which provides a per ad performance ratio. Platforms having a greater CTR are those that have more receptive audiences to the advertisements of David Jones. The CMO ought to direct creative assets to high CTR channels and look at the ad strategy on low CTR channels.

1.10 Business concern 4

Name and Description:

Customer: Which customers spend the most?

A low percentage of customers will often contribute an unfairly large portion of the revenue. This identification of such high value customers will allow the CMO to develop targeted VIP marketing, offers and loyalty programs to maintain these customers.

SQL Query

```sql
-- QUERY 4: Which customers spend the most?
SELECT c.CustomerID, c.Name, c.Location,
COUNT(o.OrderID) AS TotalOrders,
SUM(o.TotalAmount) AS TotalSpent
FROM Customer c
JOIN `Order` o ON c.CustomerID = o.CustomerID
GROUP BY c.CustomerID, c.Name, c.Location
ORDER BY TotalSpent DESC
LIMIT 10;
```

SQL Result

CustomerID	Name	Location	TotalOrders	TotalSpent
1010	Mason Harris	Sydney	2	1530.00
1020	Benjamin Baker	Brisbane	2	935.00
1019	Evelyn Adams	Sydney	1	750.00
1013	Charlotte Walker	Sydney	1	615.00
1015	Amelia Young	Brisbane	2	610.00
1001	Emma Wilson	Sydney	2	570.00
1006	Noah Taylor	Adelaide	1	560.00
1025	Sofia Roberts	Perth	1	495.00
1016	Henry King	Sydney	1	480.00
1022	Daniel Carter	Melbourne	1	445.00

Justification and Insights

This query aggregates the order information per customer and that the orders are ranked by the amount spent. The LIMIT 10 statements will only give the tip ten highest spending customers. Location is an additional attribute that would enable the CMO to discover geographic clusters of high-value customers. David Jones ought to focus on these customers by sending them personalised email messages, prioritizing them in the EOFY sale, and giving them special discount codes.

1.11 Business concern 5

Name and Description:

Products: Which products should be featured in social media campaigns?

The CMO should be aware of the most popular to be conspicuously displayed in social media advertising. Advertising low demand products wastes on advertising costs and advertising high demand products will result in increased conversions and revenue.

SQL Query

```
SELECT p.ProductName, p.Price,
SUM(c.Quantity) AS TotalUnitsSold,
SUM(c.Quantity * p.Price) AS TotalRevenu
FROM Product p
JOIN Contain c ON p.ProductID = c.Produc
GROUP BY p.ProductName, p.Price
ORDER BY TotalUnitsSold DESC;
```

SQL Result

ProductName	Price	TotalUnitsSo...	TotalRevenue
Handbag	249.99	8	1999.92
Dress Shirt	79.99	7	559.93
Sneakers	129.99	7	909.93
Sunglasses	89.99	7	629.93
Perfume	119.99	7	839.93
Casual Dress	109.99	6	659.94
Polo Shirt	69.99	6	419.94
Winter Jacket	189.99	5	949.95
Wool Coat	299.99	5	1499.95
Leather Belt	49.99	4	199.96
Denim Jeans	99.99	4	399.96
Silk Scarf	59.99	3	179.97
Running Sh...	149.99	3	449.97
Ankle Boots	179.99	3	539.97
Gym Bag	79.99	2	159.98

Justification and Insights

This query joins the Product and Contain tables to compute the total units sold and total revenue per product. Multiplying quantity by price gives an accurate revenue figure per product line. Handbags, Winter Jackets and Wool Coats are among the products that will be doing well in the EOFY sale. These best sellers should be included in Instagram and Facebook advertisements by the CMO. Low performing products should be discounted instead of being advertised to enhance effectiveness of the campaign.

1.12 Business Concern 6

Name and Description:

Revenue: Which locations generate the most revenue?

The amount of customer spending differs greatly among various cities in Australia. Knowing which cities have the highest returns can help the CMO focus geographically oriented advertising and can also promote to the local audience

SQL Query:

```
-- QUERY 6: Which city generates the most revenue?
SELECT c.Location, COUNT(o.OrderID) AS TotalOrders,
SUM(o.TotalAmount) AS TotalRevenue
FROM Customer c
JOIN `Order` o ON c.CustomerID = o.CustomerID
GROUP BY c.Location
ORDER BY TotalRevenue DESC;
```

SQL Result:

Location	TotalOrders	TotalRevenue
Sydney	12	4945.00
Brisbane	6	2115.00
Perth	4	1625.00
Melbourne	5	1450.00
Adelaide	3	1030.00

Justification and Insights

This query group customer orders according to city to display geographic revenue trends. Sydney will be the one to achieve maximum revenue and then Melbourne and Brisbane. The CMO must set aside a higher proportion of the digital advertising bidget to audience in Sydney and Melbourne. In case of low performing cities like Adelaide and Perth, the promotion needs to be targeted to demand. Example: free delivery, city specific discount.

1.13 Business Concern 7

Name and Description:

Age Group: Which age group clicks and buys the most?

The age group reacts diversely towards social media advertisement. Knowing the age group that clicks on the ads the most frequently and also transforms into a purchase will allow the CMO to refine the audience targeting settings on each social media platform.

SQL Query:

```sql
-- QUERY 7: Which age group clicks and buys the most?
SELECT
CASE
    WHEN c.Age BETWEEN 18 AND 25 THEN '18-25'
    WHEN c.Age BETWEEN 26 AND 35 THEN '26-35'
    WHEN c.Age BETWEEN 36 AND 45 THEN '36-45'
    ELSE '46+'
END AS AgeGroup,
COUNT(cl.ClickID) AS TotalClicks,
COUNT(o.OrderID) AS TotalOrders
FROM Customer c
JOIN Click cl ON c.CustomerID = cl.CustomerID
LEFT JOIN `Order` o ON c.CustomerID = o.CustomerID
GROUP BY AgeGroup
ORDER BY TotalClicks DESC;
```

SQL Result:

AgeGroup	TotalClicks	TotalOrders
26-35	22	22
36-45	13	13
18-25	5	5

Justification and Insights

The CASE statement divides the customer into the four age groups. Orders need a Left JOIN to keep customers due to clicking but not purchasing and identifying age groups with high click rates but low conversions. The age groups of the 26-35 years is anticipated to yield the highest number of clicks and purchases. To make this age group a priority, David Jones should set up Instagram and Facebook targeting. In the case of the 18- 25 group, Tik Tok and Snapchat might be a better platform.

1.14 Conclusion

The SQL queries that have been developed in this report give David Jones useful and practical insights in several aspects of its social media marketing activities. The analysis includes platforms performance, campaign return on investment, click through rate effectiveness, best-selling product analysis, geographic revenue patterns and customer age group behaviour.

The most key findings from the analysis are as follows:

- Instagram and Facebook are anticipated to perform best in the advertising of David Jones and should be given the biggest portion of the advertising Budget.
- The EOFY sale and Black Friday campaign will be expected to provide the highest ROI, and their approaches should be replicated in the future planning of campaign.
- VIP loyalty programs, personalised deals and early access to sales should be offered to high value customers who are concentrated in Sydney and Melbourne.
- Prominently featured in social media advertisements should be best selling products like Handbags, Winter Jackets and wool coats whereas poor performing products are to be discounted instead od being advertised.
- Sydney is the most profitable and should be given the biggest geographic portion of the advertising budget.
- 26 to 35 age group produces the highest number of clicks and purchases and must be the primary focus in social media audience targeting settings on all social media platforms.

By embedding these data driven analytical processes into ongoing marketing operations, David Jones can respond promptly to the changing consumer behaviour, maximize its allocation of advertising budgets, and hold a competitive edge in the Australian retail market, The database developed and deployed in this assignment can be used as a scalable base to support continuous marketing analytics and reporting.